Harsh Singh Rawat

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Profile

- Software engineer skilled in **Python, C++, SQL, Linux**, trading/risk systems, financial data pipelines, and deployment infrastructure. Strong DSA background, with exposure to PnL attribution, risk reporting, and fund accounting.

Education

Indian Institute of Information Technology Kalyani, India — Dec. 2021 - Present
Bachelor of Technology in Electronics and Communication Engineering — CGPA: 8.31/10.0

Stella Maris Inter College, Lucknow, India — Apr. 2018 - Apr. 2020
Higher Secondary (Science - PCM) — Result: 94%

Experience

AlphaGrep — Jan. 2025 - Present
Analyst (Python Developer) — Gift City, India

- **UK Fund Accounting:**
  - Built accounting pipelines for UK funds covering cash flows, PnL attribution, forwards, GSECs, and withholding tax using **Python** and **SQL**.
  - Built a **Flask** fund backend for GSECs, forwards, and cash entries, with PnL recon views orchestrated via **Airflow DAGs**.
- **Deployer:**
  - Built a **Python/TCP** deployment server managing binary and config releases across hundreds of machines.
  - Built deployment workflows handling Git pull/commit, async binary transfer, **MongoDB** updates, and Slack alerts.
  - Built a cron-based rollout scheduler with **Redis** deduplication for safe, incremental deploys.
- **CMS (Config Management System):**
  - Built a visual Query Builder UI for CMS config changes, letting users filter, edit, and delete entries without SQL.
  - Built a Preview-and-Commit flow for non-binary releases with bulk edits and row-diff highlighting.
- **RMS (Risk Management System):**
  - Built risk analytics modules using **Python** and finance metrics.
  - Automated daily/monthly risk reporting and PDF generation (**ReportLab, Dash**).

BNKM India Private Limited — May 2024 - Nov. 2024
Associate Software Developer — India

- Helped implement an **ERP** system to streamline manufacturing and inventory tracking.
- Built **API** integrations linking shop floor systems with the e-commerce platform.
- Supported dashboards and predictive maintenance systems that reduced equipment downtime.

Projects

FEC Scheme Identification
Python, TensorFlow, Keras, Sionna, Commpy, RNN, Matlab — GitHub

- Built a blind classifier using a **7-layer RNN** (3 LSTM + 4 Dense) to identify FEC schemes from demodulated signals.

Bitcoin Block Miner
Python, hashlib, ECDSA, Crypto, Linux — GitHub

- Built a Bitcoin consensus simulator for block construction via transaction filtering, script evaluation, and Merkle root calculation.

Achievements

- **Smart India Hackathon 2023:** Finalist, top **2%** among **44,000**+ teams nationally.
- **Summer of Bitcoin 2023:** Successfully completed the program as a selected participant, gaining hands-on experience in Bitcoin and open-source development.
- **e-Yantra Robotics Competition 2023:** Semi-Finalist, represented the institute in a 4-person team.
- **Codeforces:** Ranked Specialist (rating 1510).
- **CodeChef:** Achieved a 3★ rating.

Technical Skills

Languages: C++, C, Python, SQL, HTML/CSS, JavaScript, Matlab

Concepts: Data Structures and Algorithms (DSA), Machine Learning/Deep Learning, Scheduling (Cron, DAGs), Data Analysis

Tools & Frameworks: Flask, MERN Stack, ERP Systems, ReportLab, Dash, Git, Linux

Infrastructure: Airflow, Redis, MongoDB, TCP/IP, Trade Finance (RMS)

Spoken Languages: English, Hindi (Native)